Do Now · retrieval

Timed Paper 2 algorithms practice

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which structure retrieves the most recently added item first?

Q2

What is the sorted-input precondition?

Q3

What moves right as a key is inserted into a sorted prefix?

Q4

What are the worst-case time complexities of linear and binary searches?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Read the task command — A complete pass means all permitted adjacent comparisons for that pass, not just one swap.

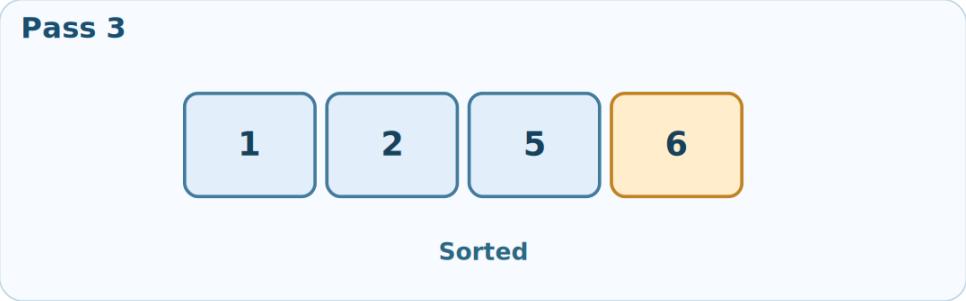
Core idea 2

Trace one pass — [6,2,5,1] becomes [2,5,1,6] after comparing each pair.

Core idea 3

Finish every pass — Continue to [2,1,5,6], then [1,2,5,6]. A clear trace shows each pass.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

trace — Record an algorithm's changing state.

sorted prefix — Part already in order during insertion sort.

precondition — Requirement, such as ordering, for an algorithm.

O(log n) — Work grows by repeated halving.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

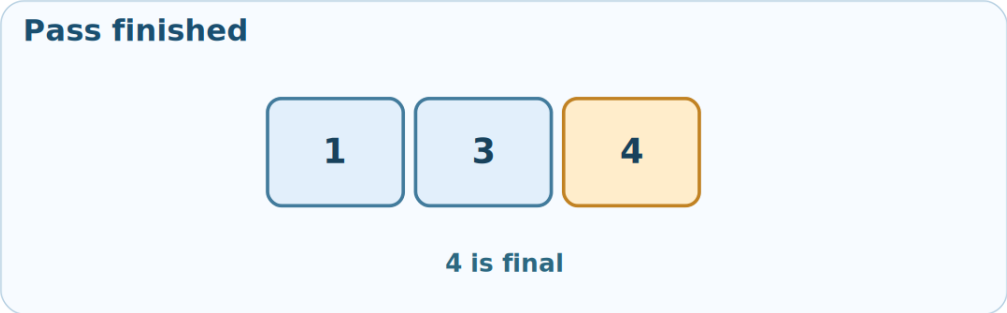
Problem

Model the first complete ascending bubble-sort pass on [4,1,3].

Follow the worked method

1. Predict first — Start [4,1,3] and compare 4 with 1: swap.
2. Check the method — Now compare 4 with 3: swap, producing [1,3,4].
3. Check the method — The pass is complete and the array is sorted.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

7 swaps with 2 giving [2,7,6,3]; then 7 swaps with 6 giving [2,6,7,3].

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

An array contains [8,3,6,1]. Trace ascending insertion sort, giving the array after each outer-loop iteration and explaining how the key is positioned.

Question 2

Write a C# function BinarySearch(int[] a,int target) returning the first matching index it encounters, or −1. Include correct bounds and a sorted-input precondition.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain one advantage and one disadvantage of merge sort over bubble sort, including time complexity, and identify an input scenario for which linear search is more practical than binary search.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Binary search works correctly on any unordered input.

Use key vocabulary from memory

1. Explain trace in your own words.

2. Explain sorted prefix in your own words.

3. Explain precondition in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.